MODULE 26 · DOMAIN 3 - COST-OPTIMIZED ARCHITECTURE DESIGN

Cost-Optimised Compute Design

Pick the cheapest compute model that fits the work, right-size it from real metrics, and scale it to the demand you actually have.

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The cheapest unit price does not help if you buy **twice the units you need**. Shape and size come before discounts.

Compute cost levers

LEVER	TYPICAL EFFECT
Graviton (Arm) instances	Better price-performance than comparable x86
Right-size from metrics	Remove unused CPU and memory
Scale with demand	Pay for peaks only while they last
Serverless for spiky work	No charge while idle
Spot or Fargate Spot	Deep discount for interruptible work

ON THE EXAM

Design levers first, purchase options second.

Step Functions: Standard or Express?

STANDARD WORKFLOWS

Long, auditable, exactly once

- Priced per state transition
- Run for up to a year
- Exactly-once execution
- Full history kept, task tokens, human waits

EXPRESS WORKFLOWS

High volume, short, cheap

- Priced per request and duration
- Run for up to five minutes
- At-least-once when asynchronous
- History sent to CloudWatch Logs

THE TIEBREAKER

Millions of short, idempotent runs: Express. Long-running, waiting or non-idempotent: Standard.

Lambda cost, decided by design

- **Cost is memory times duration**, plus requests; CPU scales with memory.
- **Arm64** is cheaper per GB-second than x86 for the same memory.
- **Tune memory:** more memory can finish faster and cost less, or not.
- **Provisioned concurrency** removes cold starts, but is paid while idle.
- **Avoid waiting:** call services asynchronously; do not pay to sleep.

> ON THE EXAM
>
> Compute Optimizer recommends Lambda memory sizes from real invocation metrics.

Lambda or Lambda Managed Instances?

LAMBDA, DEFAULT

Pay per invocation duration

- Scales to zero
- One request per environment at a time
- Ideal for spiky or low volume
- Compute Savings Plans apply

LAMBDA MANAGED INSTANCES

Lambda on EC2 you choose

- EC2 price plus a 15% management fee
- No per-request duration charge
- Many requests per environment
- Commitments apply to the EC2 part

THE TIEBREAKER

Steady, very high-volume functions that want EC2 pricing and commitments: Lambda Managed Instances.

AWS Compute Optimizer

RECOMMENDS	FOR
Instance type and size	EC2 instances and Auto Scaling groups
Volume type, size, IOPS	EBS volumes, RDS storage
Memory size	Lambda functions
CPU and memory	ECS services on Fargate
Instance class, I/O-Optimized	RDS and Aurora
Idle resources to remove	NAT gateways, caches, SageMaker endpoints, ASGs

ON THE EXAM

Rightsizing from measured utilisation, across EC2, Lambda, Fargate, EBS and RDS: Compute Optimizer.

Scaling policies and when they fit

POLICY	FITS
Target tracking	Keep a metric near a target, such as 50% CPU
Step scaling	Different responses for different breach sizes
Scheduled scaling	Known patterns, such as office hours
Predictive scaling	Recurring daily or weekly cycles, scale ahead
Fargate Spot	Interruptible ECS tasks at a large discount

ON THE EXAM

Known hours: scheduled. Recurring cycles: predictive. Everything else: target tracking.

Reading the scenario

WHEN THE QUESTION SAYS	IT IS TESTING
Millions of short, idempotent workflow runs	Step Functions Express
Portable code on x86 instances	Move to Graviton
Function memory far above what it uses	Compute Optimizer memory sizing
Steady, huge Lambda volume	Lambda Managed Instances
Idle nights and weekends	Scheduled scaling
Interruptible ECS tasks	Fargate Spot capacity provider

ON THE EXAM

Find the waste first: idle time, oversize, wrong model. Then pick the lever.

> Meridian Freight validates each tracking event with a five-step workflow that finishes in under two seconds. It runs 40 million times a day on Step Functions Standard, and every step is idempotent.
>
> The workflow bill is now larger than the compute it orchestrates.

What should the solutions architect recommend?

A Move the workflow to EC2 instances running a custom orchestrator

B Change the workflow to a Step Functions Express workflow

C Buy a Compute Savings Plan

D Add more steps to batch work inside each state

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Custom orchestrator

Replaces a managed service with code and servers to run.

B Express workflow

Per-request and duration pricing fits short, high-volume, idempotent runs.

C Compute Savings Plan

Does not cover Step Functions state transitions.

D More steps

More transitions means a bigger Standard bill.

THE SIGNAL

Short, idempotent, tens of millions of runs: Express, not Standard.

> Meridian Freight runs a Java 21 microservice estate on 300 x86 general-purpose EC2 instances. The container images are already built for both x86 and Arm.
>
> The team wants to reduce compute cost without reducing performance, before buying any new commitments.

What should the solutions architect recommend first?

A. Move to larger x86 instances and run fewer of them

B. Buy three-year Standard Reserved Instances for the current types

C. Run all services on Spot Instances

D. Migrate the services to equivalent Graviton instance types

◆ PAUSE AND DECIDE

What should the solutions architect recommend first?

A Larger x86 instances

Same price-performance, just packaged differently.

B 3-year RIs on current types

Locks in the more expensive architecture, and is a commitment.

C All Spot

Interruption risk for services not designed for it.

D Graviton

Better price-performance; images already support Arm.

THE SIGNAL

Portable code, multi-architecture images, before commitments: move to Graviton.

> A Meridian Freight Lambda function that calls partner APIs is configured with 3008 MB of memory on x86. It uses under 200 MB and spends most of its time waiting for responses.
>
> The team wants to cut its cost with minimal effort.

What should the solutions architect recommend?

A. Add provisioned concurrency

B. Apply Compute Optimizer's memory recommendation and switch the function to arm64

C. Increase memory so the function runs faster

D. Rewrite the function to run on EC2

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Provisioned concurrency

Adds an always-on charge; cold starts are not the problem.

B Right-size memory, use arm64

Pays for far less memory, at a lower per-GB-second rate.

C More memory

Waiting on the network does not get faster with more CPU.

D Rewrite for EC2

High effort, and pays for idle servers.

THE SIGNAL

I/O-bound and oversized: reduce memory. Portable runtime: arm64.

> Meridian Freight's telemetry ingestion function handles a steady 20,000 requests a second, all day, every day. Requests are short and the code is thread-safe.
>
> Finance wants to apply the company's existing commitment discounts to this workload and lower its cost, while keeping the Lambda programming model.

What should the solutions architect recommend?

A. Provisioned concurrency for the peak level

B. Rewrite the service for Amazon ECS on EC2

C. Increase the function's memory to reduce duration

D. Run the function on Lambda Managed Instances, covered by the existing Savings Plans

◆ PAUSE AND DECIDE

What should the solutions architect recommend?

A Provisioned concurrency

Addresses latency, not the cost of steady volume.

B Rewrite for ECS

Abandons the Lambda model the team wants to keep.

C More memory

Short requests gain little; cost per request may rise.

D Lambda Managed Instances

EC2 pricing with commitments, many requests per environment, same model.

THE SIGNAL

Steady, huge, thread-safe, keep Lambda, use commitments: Lambda Managed Instances.

> Meridian Freight's internal dispatch portal runs on an Auto Scaling group sized for its weekday peak around the clock. Usage is heavy from 7 a.m. to 7 p.m. on weekdays and near zero otherwise.
>
> During the day, load varies with shipment volume.

What should the solutions architect configure?

A. Scheduled actions that reduce the group to a minimum outside working hours and restore it each morning, with target tracking during the day

B. A fixed group size equal to the daily peak

C. Step scaling only, on CPU utilisation

D. Three-year Reserved Instances for the peak size

◆ PAUSE AND DECIDE

What should the solutions architect configure?

A Scheduled plus target tracking

Removes idle hours; tracks the daytime variation.

B Fixed at peak

This is the current waste.

C Step scaling only

Reacts, but keeps a large minimum unless schedules change it.

D 3-year RIs at peak

Commits to capacity that is idle most of the week.

THE SIGNAL

Known hours: scheduled actions. Variation within them: target tracking. They combine.

> Meridian Freight runs nightly report generation and all development environments as ECS services on Fargate. Tasks can be interrupted and retried without harm.
>
> The team wants to cut cost while keeping a small amount of capacity that is never interrupted.

What should the solutions architect configure?

A. Move the tasks to Lambda

B. Buy Reserved Instances for Fargate

C. A capacity provider strategy with a small base on Fargate and the rest weighted to Fargate Spot

D. Run all tasks on Fargate On-Demand with larger task sizes

◆ PAUSE AND DECIDE

What should the solutions architect configure?

A Lambda

Reports may exceed 15 minutes; a redesign for no stated need.

B Fargate RIs

Fargate is discounted by Compute Savings Plans, not RIs.

C Base plus Fargate Spot

Deep discount on interruptible tasks, with a guaranteed base.

D Larger On-Demand tasks

Costs more, not less.

THE SIGNAL

Interruptible ECS tasks with a protected minimum: capacity provider base plus Fargate Spot.

What to carry out of this module

- **Design before discounts:** model, size and scale, then commit to the floor.
- **Graviton** for portable code; **Express** for short, high-volume workflows.
- **Lambda:** right-size memory, arm64, Managed Instances for steady volume.
- **Compute Optimizer** for rightsizing and idle resources.
- **Scheduled, predictive, target tracking**, and Fargate Spot with a base.

> ON THE EXAM
>
> Name the waste, choose the lever, and commit only to what remains.